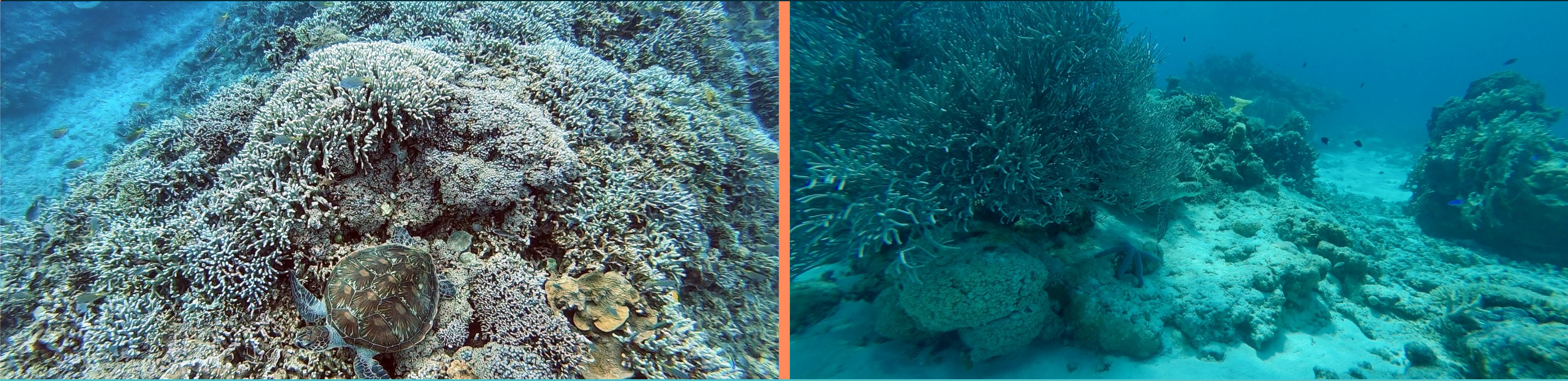


How do reef fish respond to a changing ocean?

A first look at hidden “jumping gene” activity in the brain and liver

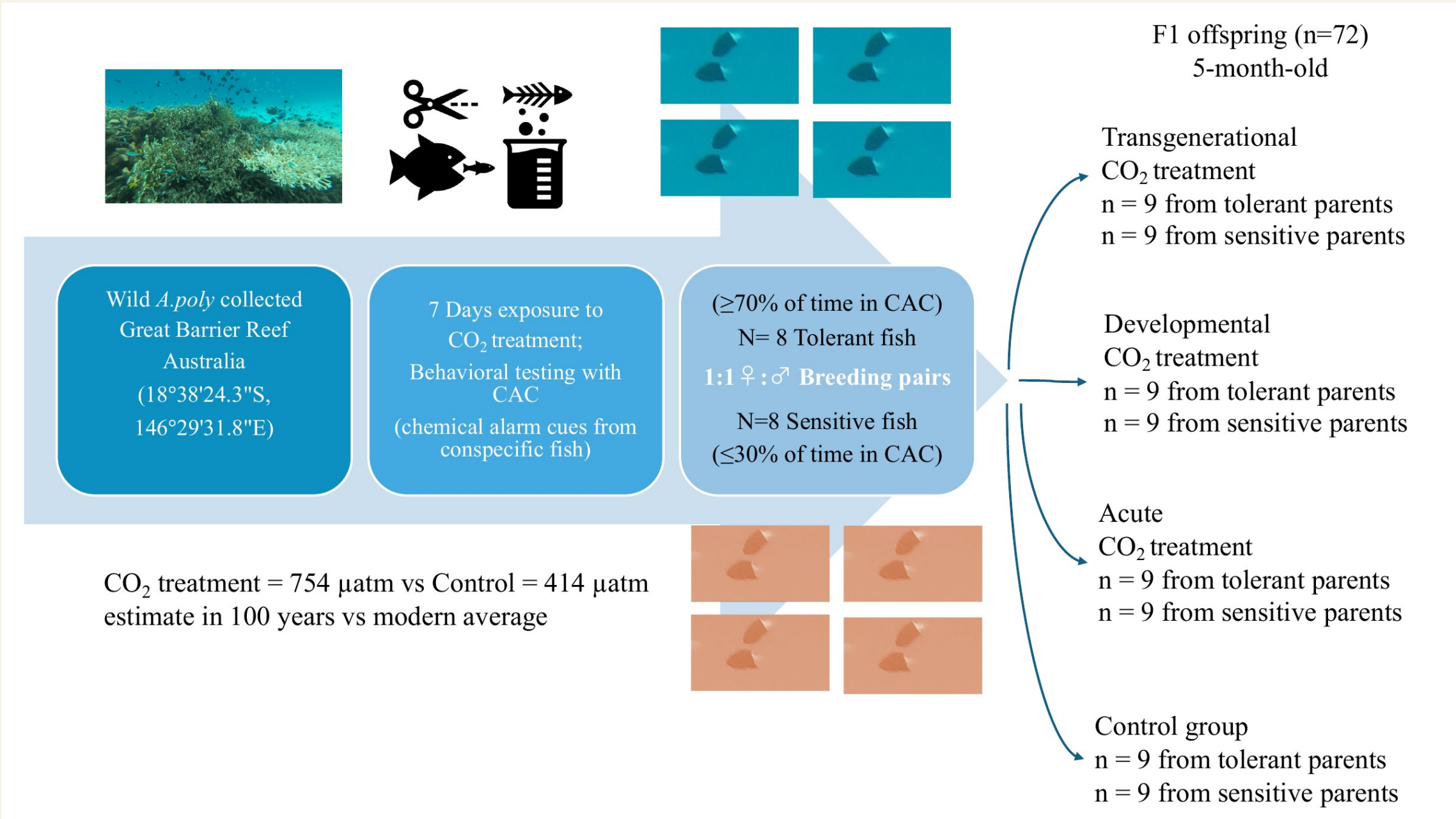
Kaisar Dauey | TE Ocean Acidification Project



Turtle and coral reef • photo supplied by Kaisar Dauey

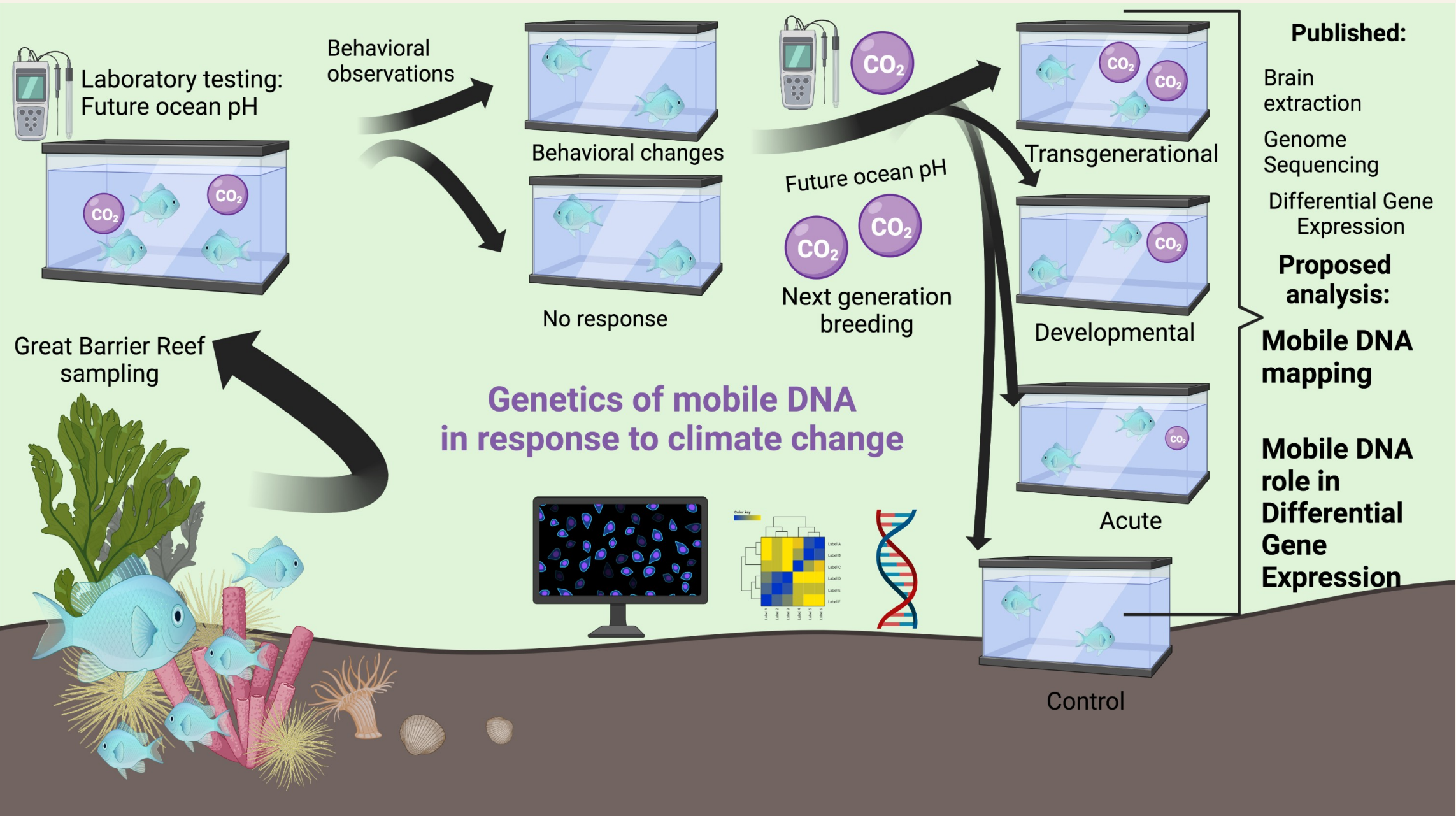
Aka Island reef, Japan • photo: Kaisar Dauey

EXPERIMENTAL DESIGN



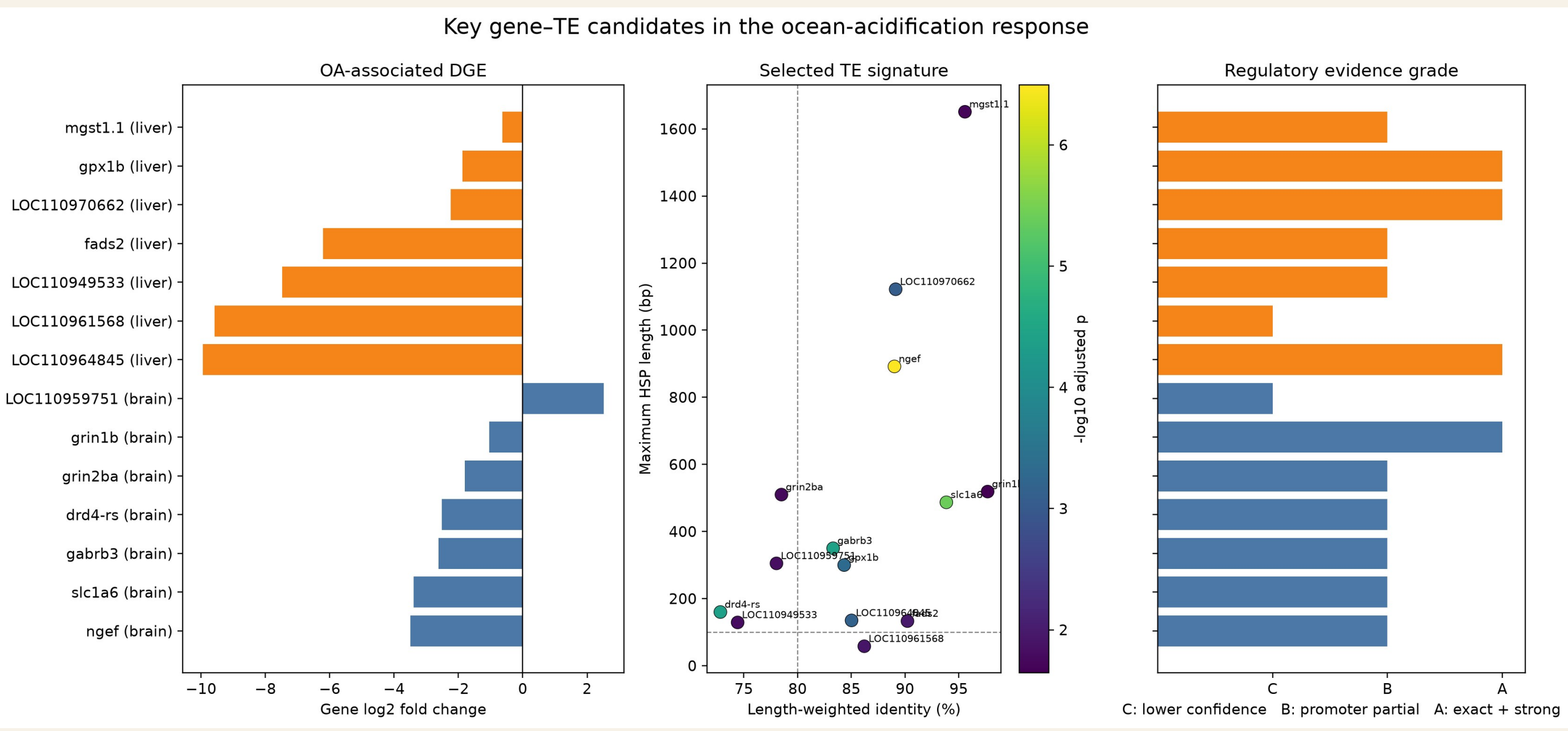
Wild parents were classified as tolerant or sensitive before offspring entered control, developmental or intergenerational CO₂ histories.

FROM REEF TO “JUMPING GENES”



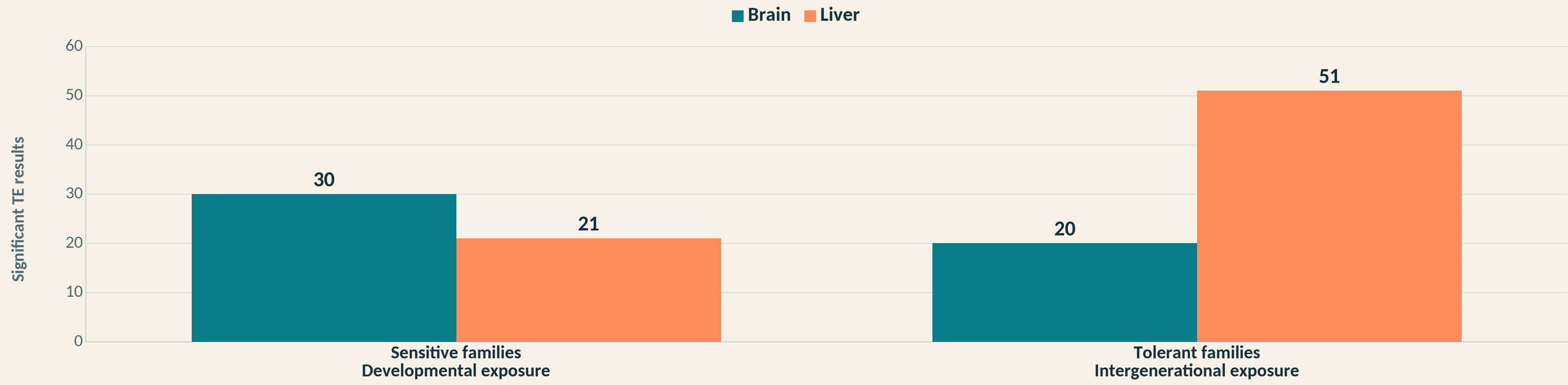
The workflow connects reef sampling and behavior with RNA sequencing, gene expression and transposable-element mapping.

GENE-EXPRESSION SIGNATURES



THE CLEAREST TISSUE PATTERN

Significant TE results in the two contrasts with the strongest brain–liver differences • FDR < 0.05



SIGNALS TO FOLLOW UP

BRAIN

gabra5 and gabrb3 point to inhibitory-neural signalling. Other prioritized candidates include ngef, grin genes and slc1a6.

LIVER

mgst1.1, gpx1b and fads2 highlight oxidative-stress, metabolism and detoxification pathways.

NEXT: VALIDATE IN 162 LIBRARIES

The larger brain–gill–liver cohort will test whether these tissue and gene signatures hold up. Association does not prove regulation.